ML Classification Project Report

ARTIFICIAL INTELLIGENCE INTERNSHIP TASK #1 • ALFIDO TECH

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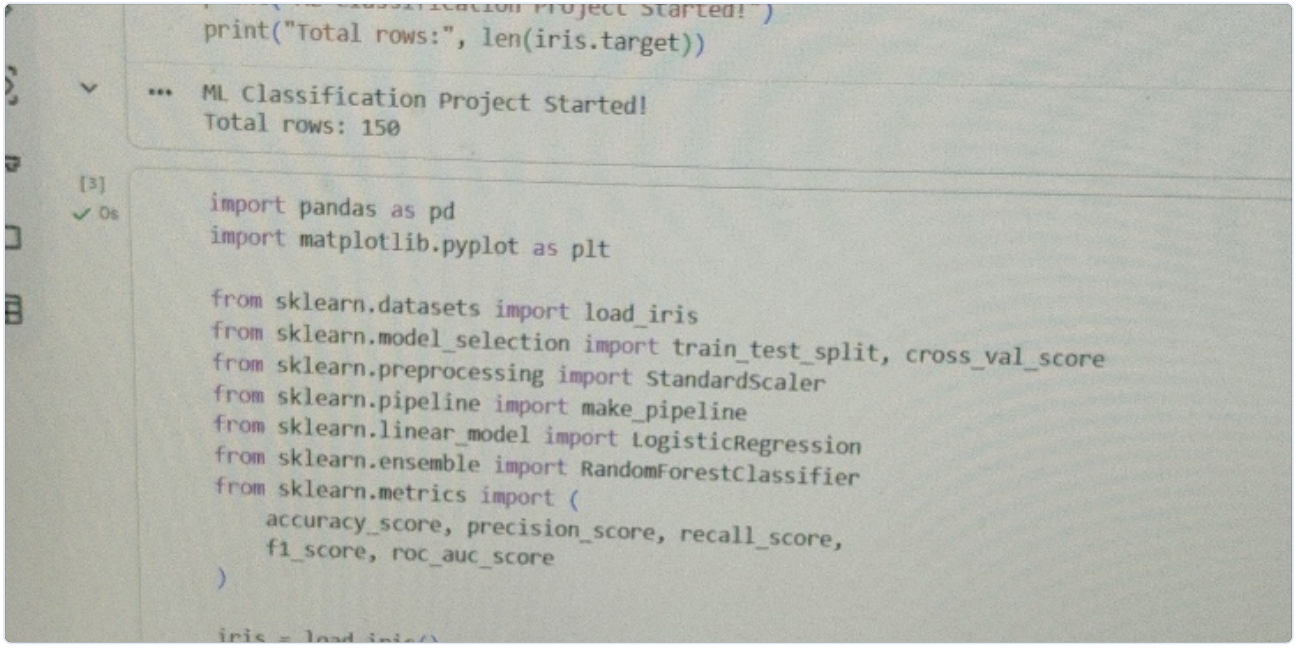
INSTITUTION
St. Mary's Women's Engineering College

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1. Project Initialization & Google Colab Workspace

The Machine Learning Classification Project was initiated in Google Colab environment (`ML_Classification_Project.ipynb`). The Iris dataset was imported via `sklearn.datasets.load_iris`, initializing 150 target instances across 4 continuous features.

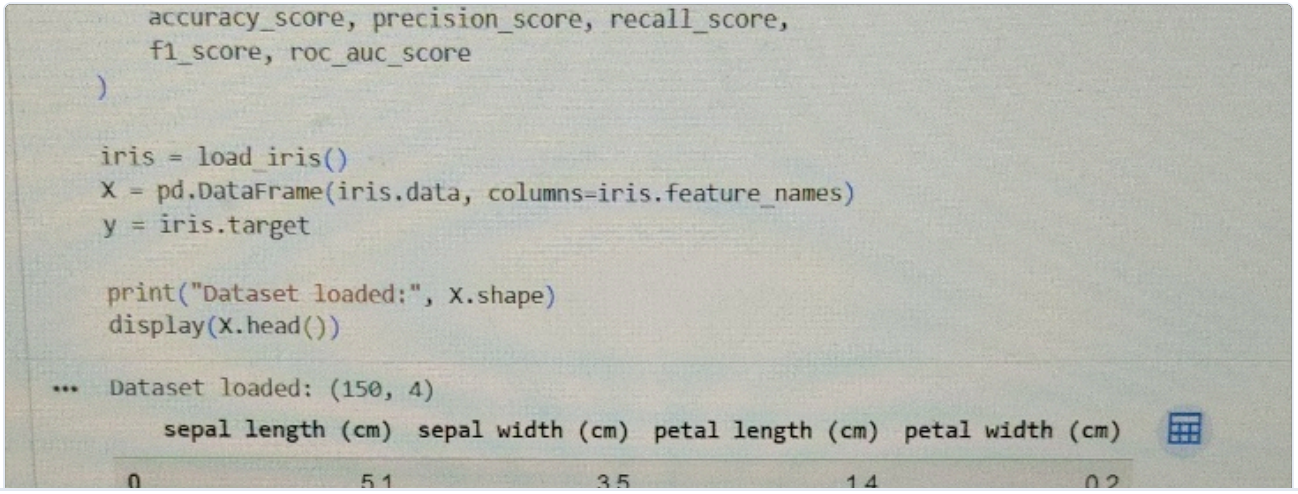
Figure 1: Notebook Environment Setup & Dataset Initialization Log Source Evidence: 216120d1



2. Feature Matrix Inspection & Data Profiling

The dataset was converted into a Pandas DataFrame to inspect feature vectors (sepal length, sepal width, petal length, petal width) and target classifications. Verification confirmed 150 clean entries with zero missing values.

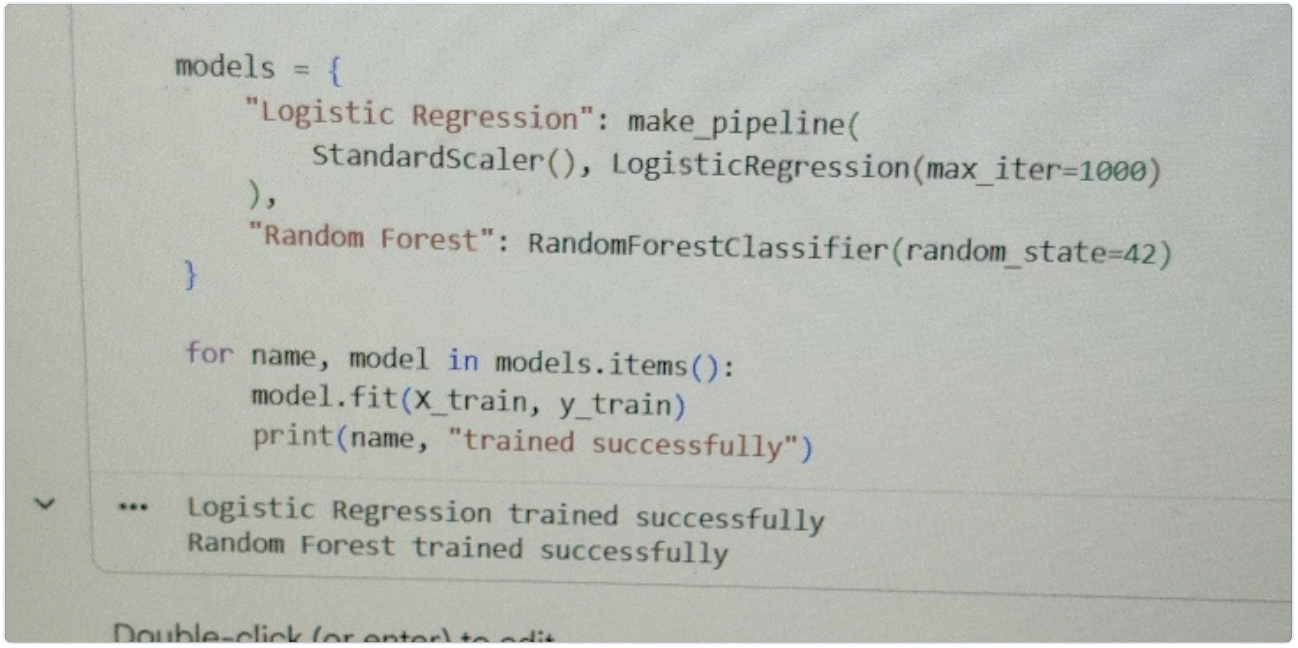
Figure 2: Pandas DataFrame Inspection (`x.shape: (150, 4)`) Source Evidence: 04792b44



3. Model Pipeline Construction & Stratified Splitting

Data was partitioned using `train_test_split` with 80/20 train-test ratio and class stratification. Logistic Regression was paired with `StandardScaler` within a pipeline, while Random Forest was initialized with `random_state=42`.

Figure 3: Stratified Train-Test Split & Model Construction Code Source Evidence: 30111d9d



```
models = {
    "Logistic Regression": make_pipeline(
        StandardScaler(), LogisticRegression(max_iter=1000)
    ),
    "Random Forest": RandomForestClassifier(random_state=42)
}

for name, model in models.items():
    model.fit(X_train, y_train)
    print(name, "trained successfully")
```

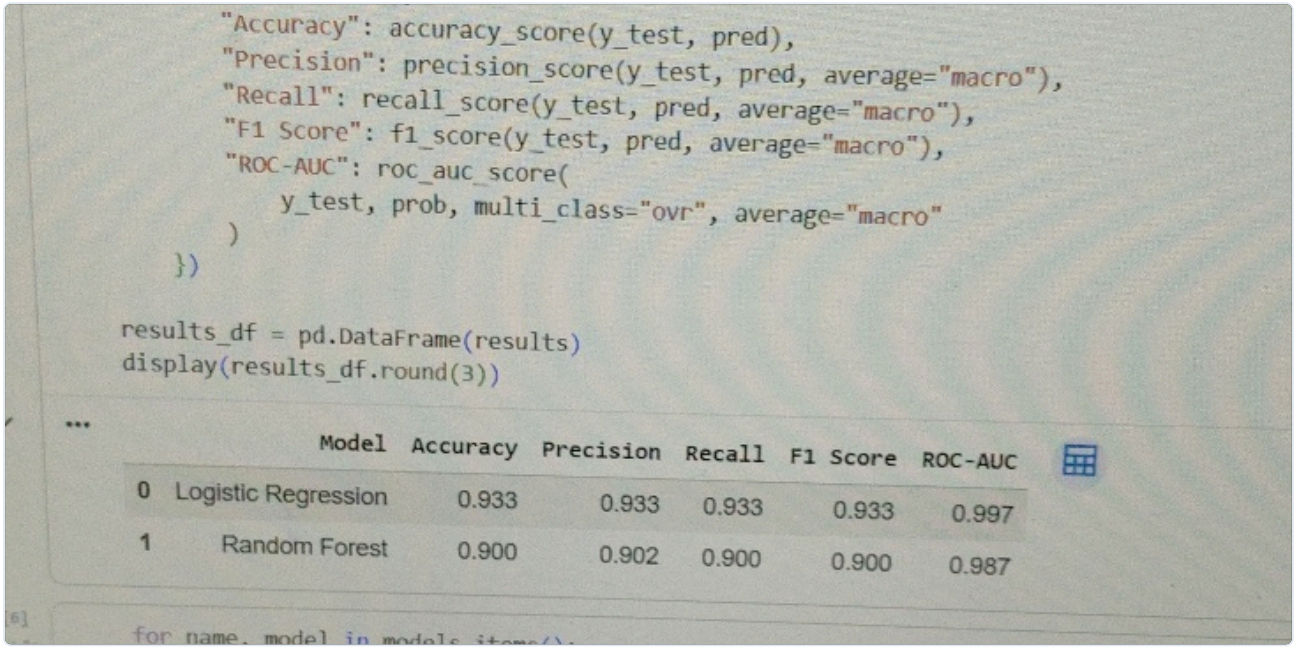
```
Logistic Regression trained successfully
Random Forest trained successfully
```

Double-click (or enter) to edit

4. Empirical Metric Evaluation & Primary Predictions

Both classifiers executed predictions on the holdout test set. Evaluated metrics included Accuracy, Macro-averaged Precision, Recall, F1-Score, and One-vs-Rest ROC-AUC.

Figure 4: Evaluated Metrics Summary DataFrame Output Source Evidence: 341b0ad4



```
        "Accuracy": accuracy_score(y_test, pred),
        "Precision": precision_score(y_test, pred, average="macro"),
        "Recall": recall_score(y_test, pred, average="macro"),
        "F1 Score": f1_score(y_test, pred, average="macro"),
        "ROC-AUC": roc_auc_score(
            y_test, prob, multi_class="ovr", average="macro"
        )
    })

results_df = pd.DataFrame(results)
display(results_df.round(3))
```

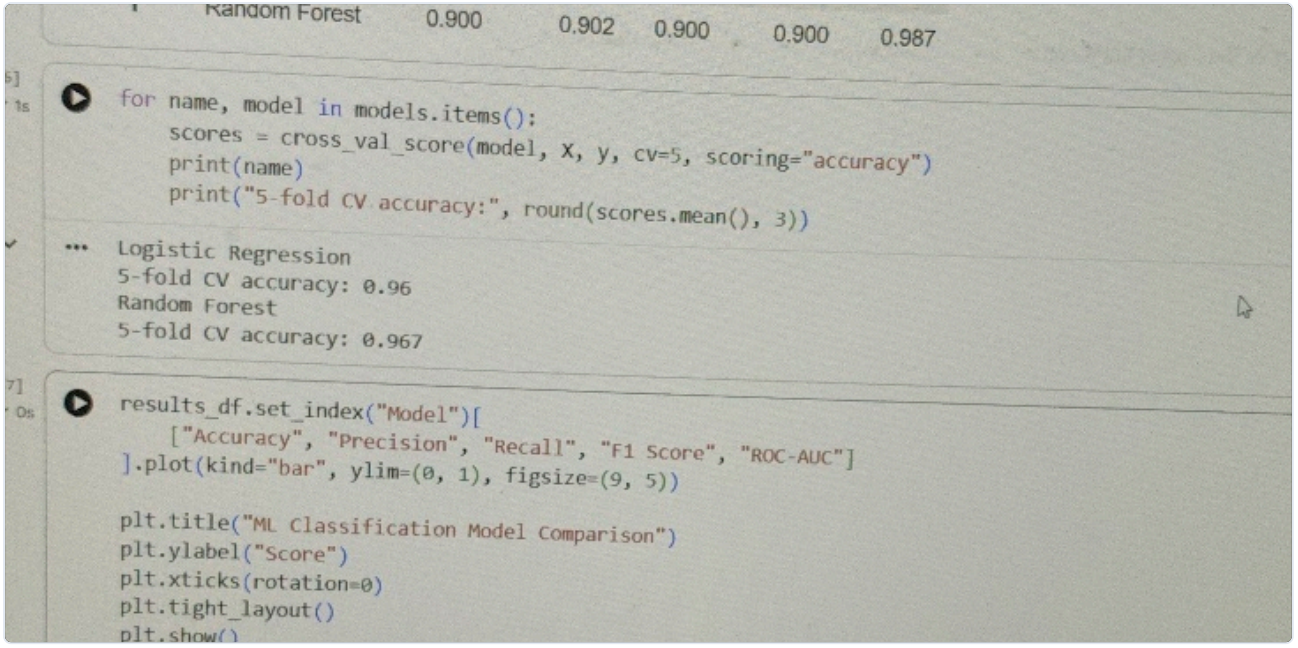
	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	Logistic Regression	0.933	0.933	0.933	0.933	0.997
1	Random Forest	0.900	0.902	0.900	0.900	0.987

for name, model in models.items():

5. 5-Fold Cross-Validation Analysis

To measure generalization stability and guard against sampling bias, 5-fold cross-validation was computed across the dataset. Random Forest demonstrated superior stability with a mean CV score of **0.967**.

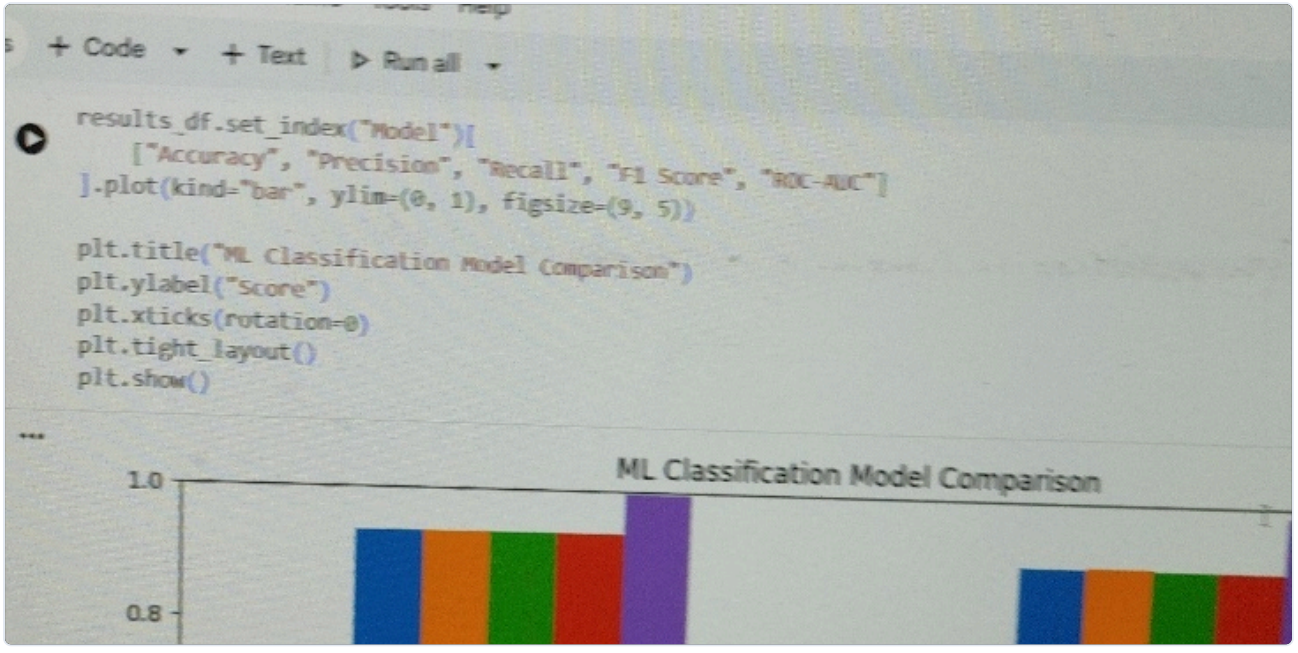
Figure 5: 5-Fold Cross-Validation Code & Execution Output Source Evidence: 9cbf1126



6. Visualization Script & Matplotlib Execution

The evaluation results were indexed and rendered into a comparative bar chart using Matplotlib, mapping performance metrics side by side for direct algorithm benchmarking.

Figure 6: Bar Plot Visualization Script & Execution Cell Source Evidence: fb6871a0

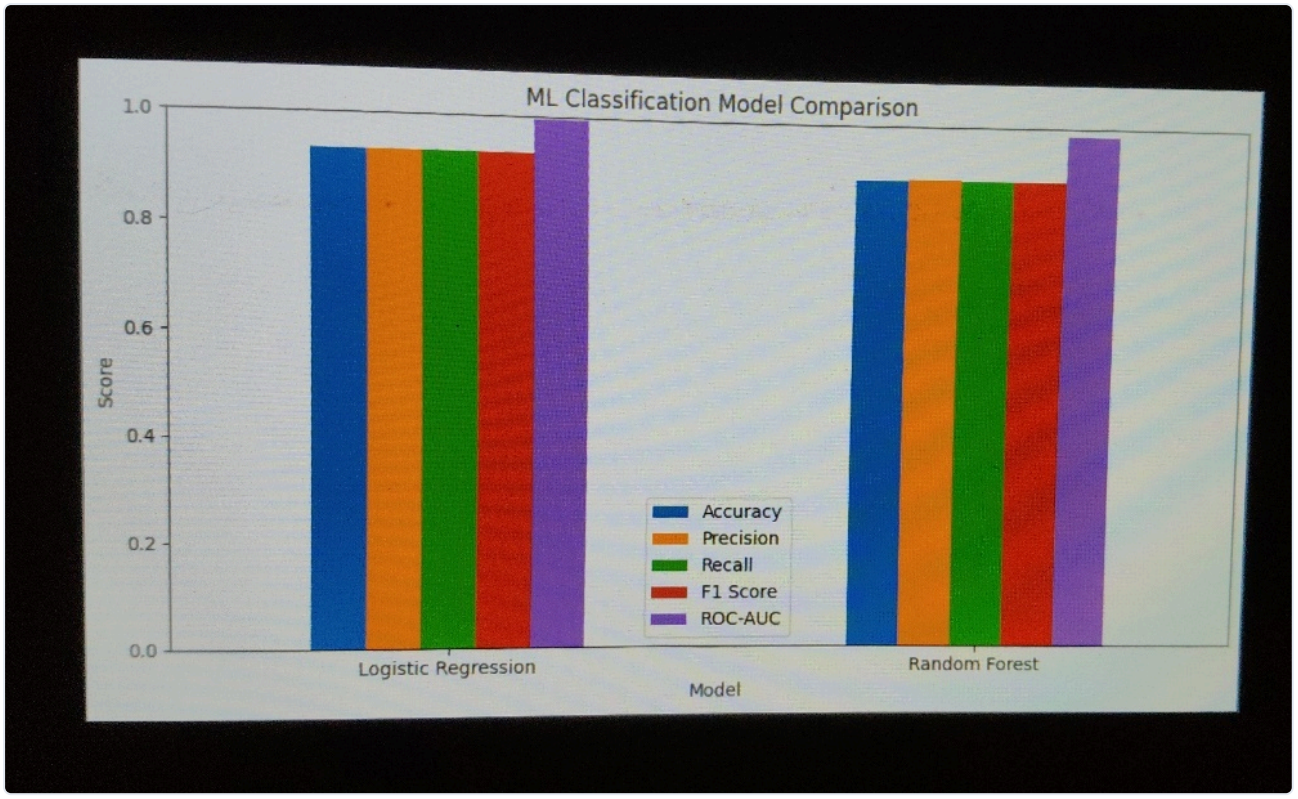


7. Benchmark Metrics & Clean Model Comparison Graph

The complete performance metric evaluation for both candidate models is summarized below, accompanied by the clean comparison bar chart.

Model Algorithm	Test Accuracy	Precision	Recall	F1-Score	ROC-AUC	5-Fold CV Score
Logistic Regression (Scaled)	**0.933**	0.933	0.933	0.933	0.997	0.960
Random Forest Classifier	0.900	0.902	0.900	0.900	0.987	**0.967**

Figure 7: Clean ML Classification Model Comparison Graph Source Evidence: 6b898e4a



8. Project Conclusion & Final Assessment

Key Findings & Summary: The Machine Learning Classification Task successfully completed model development, validation, and benchmarking.

- **Primary Accuracy:** Scaled Logistic Regression achieved a primary test accuracy of **93.3%** with an impressive ROC-AUC of **0.997**, confirming strong linear separability across flower categories.
- **Cross-Validation Stability:** Random Forest Classifier demonstrated superior cross-validation stability with a mean 5-fold accuracy of **96.7%** (compared to 96.0% for Logistic Regression).
- **Final Deployment Verdict:** Logistic Regression is recommended for deployment scenarios prioritizing maximum interpretability and low execution latency, while Random Forest is preferred for robustness across high-variance datasets.

REPORT PREPARED BY
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